MR3 Communication Protocol

Com Channel Settings

- Baud rate: 115200
- Data Bits: 8
- Stop Bits: 1
- Parity: None

Communication with the meter follows a general rule: 6 bytes sent from the controlling device will initiate a command. To retrieve data:

Send 6 Bytes:

To get next sample: 6 x – 0x03

To set Time to zero (and get next sample): 6 x – 0x04

Receive 31 Bytes:

6 – bytes for Time

6 – bytes for X Axis

6 – bytes for Y Axis

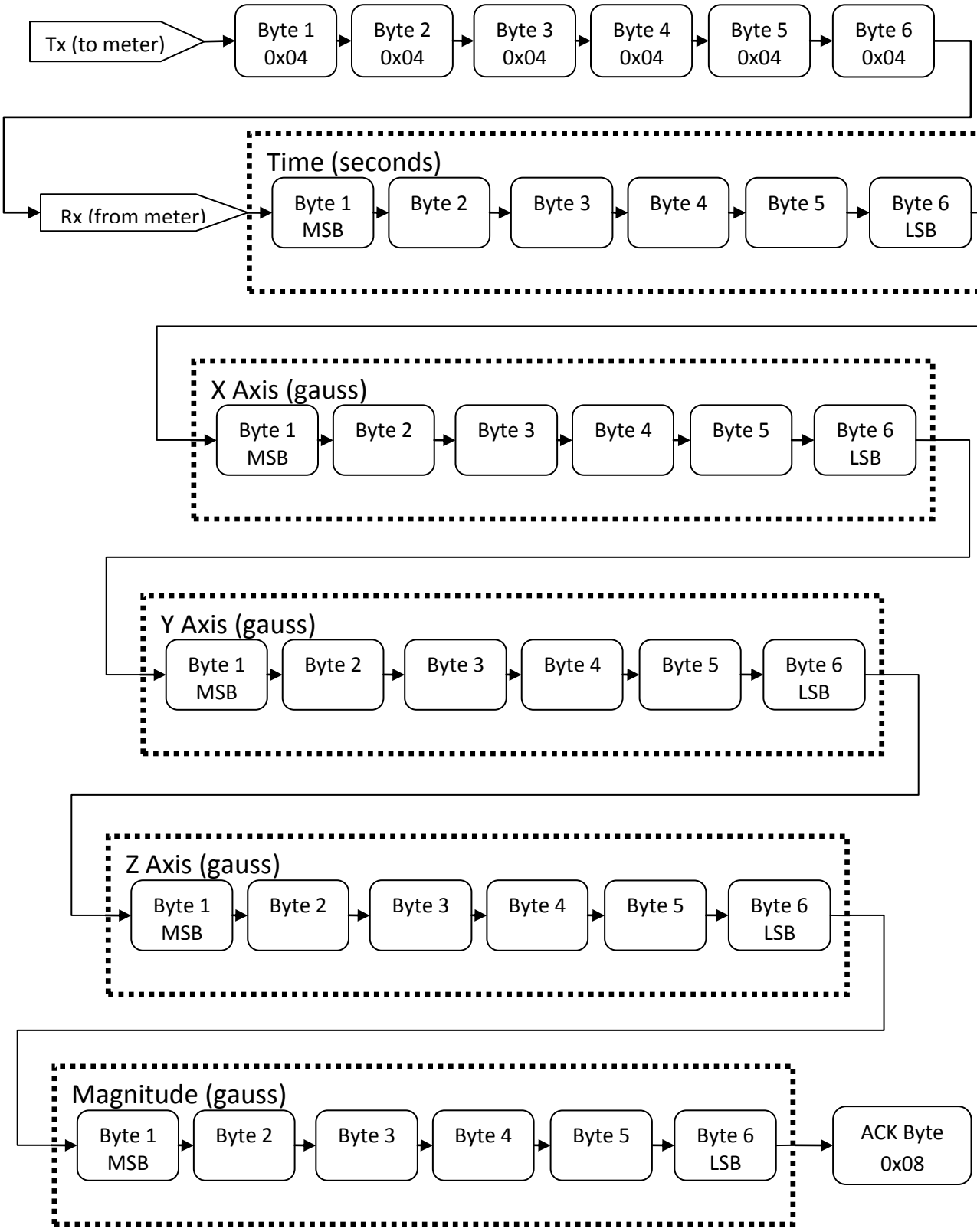
6 – bytes for Z Axis

6 – bytes for Magnitude

1 – byte for Acknowledgment (should be 0x08)

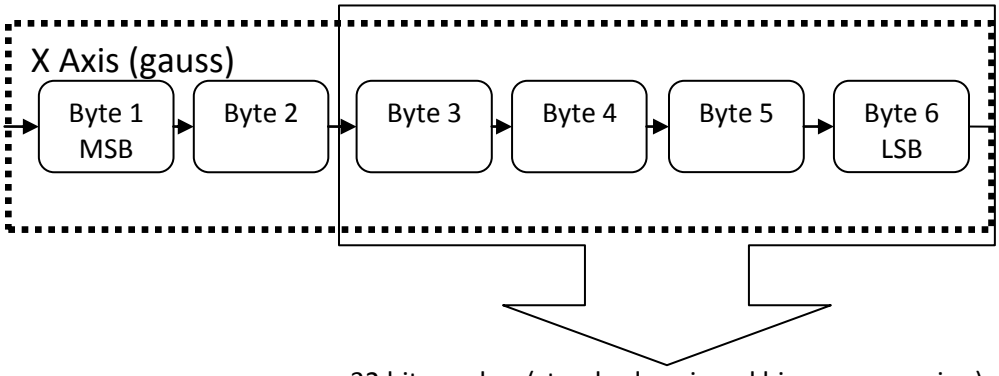
Data Stream Example

(streaming data from MR3 - with time set to zero)



Data Stream Decomposition

(convert a single axis)

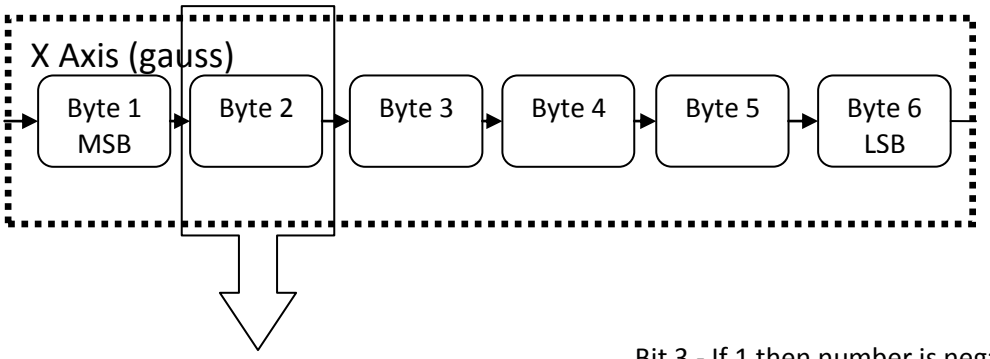


32 bit number (standard unsigned binary conversion)

Example:

Byte 3	Byte 4	Byte 5	Byte 6
00000000	00000000	00000001	11110110

Becomes: 502



Holds sign and decimal information

x	x	x	x	+/-	D	D	D

Bit 7 MSB — Bit 0 LSB

Bit 3 - If 1 then number is negative
- If 0 then number is positive

Bit 2 - 0 - Decimal register; convert to number (0 - 7) - decimal is this many places from right

Example:

Byte 2	Byte 3	Byte 4	Byte 5	Byte 6
00001010	00000000	00000000	00000001	11110110

Becomes: -5.02